

On the Mandelbrot Set

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1 Introduction and Motivation

During our time studying complex-valued functions in this course, we have seen how we can define such strong restrictions on functions such as holomorphicity. In contrast to this, one of the remarkable elements of the Mandelbrot set is how its simple construction can lead to such complex and beautiful results. In this report, we will discover the mathematics governing the Mandelbrot and Julia set's construction, and how their geometry can be linked to the Fibonacci sequence.

2 Construction of The Mandelbrot Set

The definition of the Mandelbrot set is surprisingly simple, so we can immediately define it as follows.

Definition 1 ([4]). Let $f : \mathbb{C} \rightarrow \mathbb{C}$ where, $f_c(z) = z^2 + c$. Then we denote the Mandelbrot set M as,

$$M = \{c \in \mathbb{C} : f_c^n(0) \text{ remains bounded as } n \rightarrow \infty\},$$

where $f_c^n(z)$ denotes that f_c has been applied n times to z .

Using computer processing we can produce images of the Mandelbrot set, such as Figure 1, which shows the complexity of the Mandelbrot set.

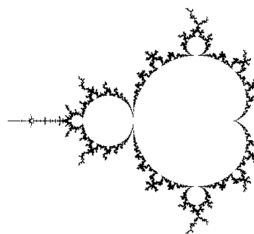


Figure 1: Mandelbrot set in $\mathbb{C}$ [6].

Now to clarify what this represents, if we have the function $f_c(z) = z^2 + c$ the Mandelbrot set in Figure 1 represents all the values of c for a starting z -value $z_0 = 0$, such that when we repeatedly apply f_c to z_0 it remains bounded. This can be regarded as the parameter space wherein the parameter is varied [1].

What if instead, we wanted to keep c constant and find starting values z_0 such that $f_c^n(z_0)$ remains bounded? Then we produce what is known as the filled Julia set [1]. To define the filled Julia set we first need to define the basin function.

Definition 2 ([7]). Let $p \in \hat{\mathbb{C}}$ then the basin of p over a function f is,

$$\text{Bas}(p) = \left\{ z \in \hat{\mathbb{C}} : f^n(z) \rightarrow p \text{ as } n \rightarrow \infty \right\},$$

where $\hat{\mathbb{C}}$ is the extended complex plane [3].

This represents all the points $z \in \mathbb{C}$ that tend towards p when a function f is repeatedly applied.

Now we can define the filled Julia set as follows.

Definition 3 ([7]). The **filled Julia set** of a function f is,

$$\mathcal{K}_f = \hat{\mathbb{C}} \setminus \text{Bas}(\infty) = \{z \in \mathbb{C} : f^n(z) \text{ is bounded}\}.$$

Now we have properly defined the Mandelbrot and Julia sets, we can arrive at more profound results such as The Fundamental Dichotomy Theorem.

Theorem 4 (5.1 [1]). *For each c , the filled Julia set is either a connected set or a Cantor set. More precisely, if the orbit of 0 escapes to infinity, that is, if $c \in M$, $\mathcal{K}_{f_c}$ is a Cantor set. If the orbit does not escape to infinity, that is, $c \notin M$, $\mathcal{K}_{f_c}$ is connected.*

Where the orbit is the sequence $z_0, z_1, \dots$ where $z_{n+1} = f_c(z_n)$ [1], and Cantor sets have been defined in the Honours Analysis course but are also defined in "An Introduction to Julia and Fatou Sets" [7].

3 Geometry of the Mandelbrot Set

To investigate the geometry of the Mandelbrot Set we will first state some important definitions.

Definition 5. A fixed point z^* of a function f , satisfies $f(z^*) = z^*$. z^* is an attracting fixed point when $|f'(z^*)| < 1$.

The obvious question now would be what are the fixed points of $f_c(z) = z^2 + c$? Let z^* be a fixed point of f_c then, $z^* = (z^*)^2 + c$. So z^* is an

attracting fixed point when $|f'_c(z^*)| < 1$. Now, if we want the boundary of this set in the parameter space then this would be when $|f'_c(z^*)| = 1$, since 1 can be expressed as $1 = e^{2\pi it}$. So we have,

$$(z^*)^2 + c = z^*$$

$$2z^* = e^{2\pi it}.$$

Solving these simultaneously for c we get that $c = \frac{e^{it}}{2} - \frac{e^{2it}}{4}$ [5] which is the parameterized formula for the main cardioid of the Mandelbrot set. Thus the fixed points of the Mandelbrot set form a cardioid.

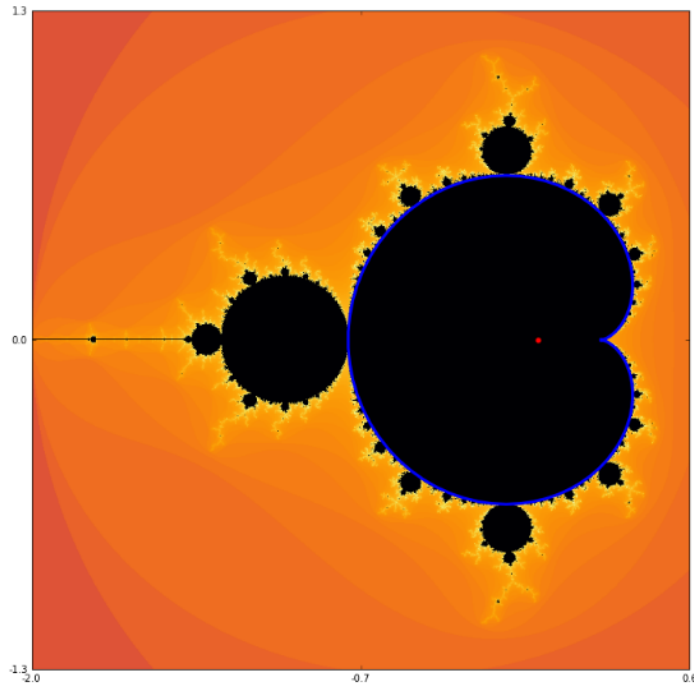


Figure 2: Highlighting the main cardioid of the Mandelbrot set [5].

4 Fibonacci in The Mandelbrot Set

We will now uncover how the Fibonacci sequence is hidden within the main cardioid of the Mandelbrot set. In Figure 2 we can see as we trace around the main cardioid there are bulb structures which branch off and form the rest of the Mandelbrot set. Each of these bulbs can be assigned a unique rotation number $\frac{p}{q}$. To determine the rotation number of a bulb, we must first introduce the period of a bulb.

Definition 6. The **period of a bulb** is the value n , such that the orbit of 0 tends to an attracting cycle of period n ($f_c^n(z_0) = z_0$) [2].

Where c is some value in the interior [3] of the bulb.

So back to the rotation number, the denominator q is simply the period of the bulb, now we are left to find p . If we deform the main cardioid into a circle and superimpose a ray from the bulb to the origin. Then the ray will make an angle with the positive real axis, that is an integer multiple of $\frac{1}{q}$. This multiple is exactly p , you can see how this manifests in Figure 3.

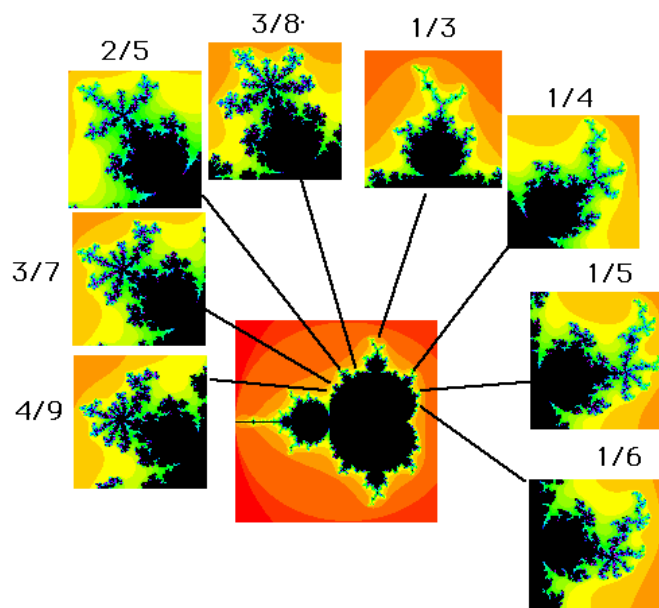


Figure 3: Some rotation numbers of bulbs [2].

Now if we were to find the largest (area within $\mathbb{C}$) bulb between the $\frac{1}{2}$ and $\frac{1}{3}$ bulbs we would find it would be the $\frac{2}{5}$ bulb. Again if we were to find the largest bulb between the $\frac{1}{3}$ and $\frac{2}{5}$ bulbs we would find it would be the $\frac{3}{8}$ bulb. Observing the period (denominator) of these bulbs, we see a pattern emerge: 2,3,5,8,... This is the Fibonacci sequence but how did it get here? We can find the largest bulb between two bulbs by taking the Farey sum ($\oplus$) of their rotation numbers [2]. Since we started with, $\frac{1}{2} \oplus \frac{1}{3} = \frac{2}{5}$, we can see that repeatedly doing this will result in the denominator following the Fibonacci sequence!

5 Conclusion

Over this report, we have seen how such simple beginnings can create complex and fruitful Mathematical objects through some investigation. The Mandelbrot set served as a reminder that Mathematics is connected at the deepest roots, as we saw the Fibonacci sequence, something that's associated with number theory, was hidden within complex dynamics.

References

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